

A 2-adic proof of the two-row $d = 4$ fiber law for all $b \equiv 1 \pmod{4}$

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Object. For $\lambda = (2m - b, b)$, $0 \leq b \leq m$, set $s = 1 + i$, $P(u) = (1 + su + u^2)^m$, and

$$I_b(m) := \text{Im}[u^b]((1 - u)P(u)) \in \mathbb{Z}.$$

A previous cycle proved that $I_b(m)$ is a polynomial in m whose only integer roots are the *forced* roots $m \in \{0, \dots, R\}$, $R := \lfloor (b - 1)/2 \rfloor$, and that, writing $(m)_{R+1} := \prod_{r=0}^R (m - r)$ and $I_b(m) = (m)_{R+1} Q_b(m)$, the two-row $d = 4$ fiber law $G_{(2m-b, b)}(i) = 0 \iff (2, 2)$ is equivalent to

$$(\star) \quad Q_b(m) \neq 0 \quad \text{for every integer } m \geq b.$$

Since $m \geq b > R$ gives $(m)_{R+1} \neq 0$, $(\star)$ is the same as $I_b(m) \neq 0$ for $m \geq b$.

1 An exact binomial expansion

Writing $1 + su + u^2 = (1 + u^2) + su$ and expanding, $P(u) = \sum_{j \geq 0} \binom{m}{j} s^j u^j (1 + u^2)^{m-j}$, so

$$I_b(m) = \sum_{j=1}^b \tau_j(m), \quad \tau_j(m) = \binom{m}{j} \text{Im}((1 + i)^j) (T(b - j) - T(b - 1 - j)), \quad (\text{E})$$

where $T(l) = [u^l](1 + u^2)^{m-j} = \binom{m-j}{l/2}$ for l even and 0 for l odd. Exactly one of $b - j, b - 1 - j$ is even, so $T(b - j) - T(b - 1 - j) = \pm \binom{m-j}{\beta_j}$, $\beta_j := \lfloor (b - j)/2 \rfloor$. Moreover $\text{Im}((1 + i)^j) = 2^{j/2} \sin(j\pi/4) \in \mathbb{Z}$, with $v_2(\text{Im}((1 + i)^j)) = \lfloor j/2 \rfloor$ for $j \not\equiv 0 \pmod{4}$ and $\text{Im}((1 + i)^j) = 0$ for $j \equiv 0 \pmod{4}$. In particular $\text{Im}((1 + i)^j)$ is *odd* iff $j = 1$.

Corollary 1 (mod 2 law). *For every b , $I_b(m) \equiv (R + 1) \binom{m}{R+1} \pmod{2}$.*

Proof. Reduce (E) mod 2; only $j = 1$ survives, and $|\tau_1(m)| = m \binom{m-1}{R} = (m)_{R+1}/R! = (R + 1) \binom{m}{R+1}$. $\square$

2 The law for $b \equiv 1 \pmod{4}$

Theorem 1. *Let $b \equiv 1 \pmod{4}$, $b \geq 5$, $R = (b - 1)/2$. Then for every integer m ,*

$$v_2(I_b(m)) = v_2((m)_{R+1}) - v_2(R!). \quad (\text{V})$$

Hence $v_2(Q_b(m)) = -v_2(R!)$ is constant, $Q_b(m) \neq 0$ for all integer m , and the two-row $d = 4$ law holds for all $b \equiv 1 \pmod{4}$.

Proof. As b is odd with $b - 1 = 2R$ even and $b - 2$ odd, the $j = 1$ term of (E) is exactly $\tau_1(m) = m \binom{m-1}{R} = (m)_{R+1}/R!$, so $v_2(\tau_1(m))$ equals the right side of (V). We show $v_2(\tau_j(m)) \geq v_2(\tau_1(m)) + 1$ for all $2 \leq j \leq b$ (those $j \equiv 0 \pmod{4}$ give $\tau_j \equiv 0$ and may be ignored).

Fix such a j and put $h := \lfloor j/2 \rfloor$, $d_j := \lfloor (j-1)/2 \rfloor$, $s_j := j + \beta_j$. For b odd one has $s_j = \lfloor (b+j)/2 \rfloor$, $s_j - j = R - h$, and $d_j = s_j - (R + 1)$. From $\binom{m}{j} \binom{m-j}{\beta_j} = \binom{m}{s_j} \binom{s_j}{j}$,

$$v_2(\tau_j(m)) = v_2 \binom{m}{s_j} + v_2 \binom{s_j}{j} + \left\lfloor \frac{j}{2} \right\rfloor.$$

Telescoping $\binom{m}{s_j} / \binom{m}{R+1} = \prod_{t=1}^{d_j} (m - R - t) / (R + 1 + t)$ and using that $R + 1 = (b + 1)/2$ is *odd* (so $v_2(\tau_1) = v_2 \binom{m}{R+1}$) gives

$$v_2(\tau_j(m)) - v_2(\tau_1(m)) = D_j + \sum_{t=1}^{d_j} v_2(m - R - t), \quad D_j := v_2 \binom{s_j}{j} + \left\lfloor \frac{j}{2} \right\rfloor - \sum_{t=1}^{d_j} v_2(R + 1 + t). \quad (\Delta)$$

With $v_2((2h)!) = v_2((2h + 1)!) = h + v_2(h!)$ and $\sum_{t=1}^{d_j} v_2(R + 1 + t) = v_2((R + 1)!) - v_2((s_j - j)!)$ (as $s_j - j = R - h$), a direct simplification collapses D_j to a product of $h + 1$ consecutive integers divided by $h!$:

$$D_j = v_2((h + 1)! \binom{R+1}{h+1}) - v_2(h!) = v_2(h + 1) + v_2 \binom{R + 1}{h + 1} \geq 0. \quad (\text{D})$$

Now combine (Δ) and (D) :

- If $D_j \geq 1$, then $v_2(\tau_j) - v_2(\tau_1) \geq 1$.
- If $D_j = 0$, then by (D) $h + 1$ is odd (h even) and $\binom{R+1}{h+1}$ is odd. Since $j \equiv 0 \pmod{4}$ is excluded, h even forces j odd, i.e. $j = 2h + 1 \equiv 1 \pmod{4}$ with $h \geq 2$ (the case $h = 0$ is $j = 1$). Then $d_j = h \geq 2$, and among the ≥ 2 consecutive integers $m - R - 1, \dots, m - R - d_j$ at least one is even, so $\sum_t v_2(m - R - t) \geq 1$ and again $v_2(\tau_j) - v_2(\tau_1) \geq 1$.

Thus the tail $\sum_{j \geq 2} \tau_j$ has $v_2 \geq v_2(\tau_1) + 1$, whence $v_2(I_b(m)) = v_2(\tau_1(m))$, which is (V). $\square$

3 The remaining classes

For $b \equiv 0 \pmod{4}$, $R + 1 = b/2$ is even, so D_j acquires a correction $-v_2(R + 1)$ and the $j = 1$ term only *ties* (never loses) in valuation; the identity (V) still holds numerically (verified $b \leq 40$, random $m \leq 3 \cdot 10^4$), the sole gap being a mod-4 refinement of the Corollary showing the valuation-minimal terms sum to a 2-adic unit. For $b \equiv 2, 3 \pmod{4}$ the monic integer part q_b of Q_b satisfies $q_b(m) \equiv m(m^2 + m + 1)^{\lfloor (b-1)/4 \rfloor} \pmod{2}$, i.e. it has a genuine root $m \equiv 0 \pmod{2}$; the 2-adic method is structurally inapplicable and an odd-prime Newton-polygon / irreducibility input is required. (Equivalently, for $b \equiv 0, 1 \pmod{4}$, $q_b \equiv (m^2 + m + 1)^{\lfloor (b-1)/4 \rfloor} \pmod{2}$, which has no root mod 2.)